

LECTURE 9

Aggregation and Grouping

Summarizing many tuples into meaningful answers

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

What This Lecture Covers

- 1 Why relational algebra's five basic operators cannot answer summary questions
- 2 The aggregate functions: COUNT, SUM, AVG, MIN, MAX
- 3 The aggregate operator $\mathcal{F}$ (script F)
- 4 The grouping operator $\mathcal{G}$ (script G) and grouping attributes
- 5 Combining grouping with aggregation to answer per-group questions
- 6 Worked examples and common pitfalls

A New Kind of Question

- σ , π , and $\bowtie$ always return a relation of the same general shape — still rows and columns, one tuple per matching input row.
- “What is the total salary bill?” is not about which tuples to keep — it needs every tuple reduced to a single number.
- No σ , π , or $\bowtie$ expression can produce a value computed across many tuples — relational algebra needs a new operator family.

Staff (staffNo, name, position, salary, branchNo)

staffNo	name	position	salary	branchNo
SL21	John White	Manager	30000	B005
SG37	Ann Beech	Assistant	12000	B003
SG14	David Ford	Supervisor	18000	B003
SA9	Mary Howe	Assistant	9000	B007
SL41	Julie Lee	Manager	22000	B005

*Sum of salary = 30000 + 12000 + 18000 + 9000 + 22000
= 91000*

Five Standard Aggregate Functions

Function	Meaning	Applies to
COUNT	Number of values (or tuples)	Any type
SUM	Total of all values	Numeric only
AVG	Arithmetic mean of all values	Numeric only
MIN	Smallest value	Numeric / orderable
MAX	Largest value	Numeric / orderable

Applied to Staff.salary

Function	Result
COUNT(salary)	5
SUM(salary)	91000
AVG(salary)	18200
MIN(salary)	9000
MAX(salary)	30000

NOTE

COUNT is the only aggregate that works on any column, including non-numeric ones, and COUNT(*) counts tuples directly, ignoring columns entirely.

NOTATION

The Aggregate Operator, $\mathcal{F}$

$\mathcal{F}$ <function-list> (R)

- $\mathcal{F}$ applies one or more aggregate functions across an entire relation.
- With no grouping attribute specified, the whole relation is treated as one group.
- The result is always a relation with exactly one tuple — no matter how many input rows.

```
 $\mathcal{F}$  SUM(salary), AVG(salary) (Staff)
```

Result

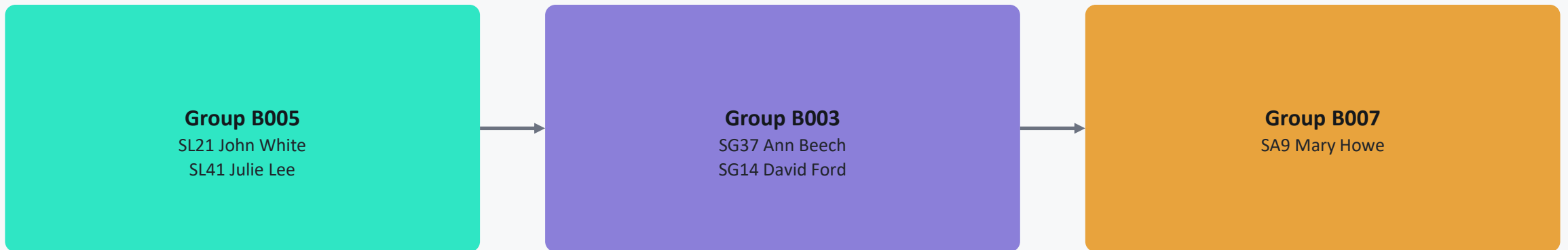
SUM(salary)	AVG(salary)
91000	18200

WARNING

Aggregation is not filtering — $\mathcal{F}$ AVG(salary)(Staff) does NOT list each staff member alongside the average. It discards every individual row and returns only the single computed number.

Grouping Splits a Relation into Buckets

- Grouping partitions a relation into groups sharing the same value of one or more grouping attributes.
- Grouping alone adds no new information — it's purely organizational, until an aggregate is applied to each group.
- Every distinct value of the grouping attribute gets exactly one group — groups are never declared in advance.



Grouping Staff by branchNo, before any aggregate is applied

The Grouping Operator, $\mathcal{G}$

```
<grouping-attributes>  $\mathcal{G}$   $\mathcal{F}$  <function-list> (R)
```

- Read left to right: partition R into groups sharing the same value of grouping-attributes, then compute the function-list separately within each group.
- The result has one tuple per group, rather than one tuple for the whole relation.

TIP

Grouping by the wrong attribute is a silent logic error, not a syntax error — the query still runs and still returns a plausible-looking table. State the question in plain English first.

WORKED EXAMPLE 1

Total Salary Per Branch

```
branchNo G F SUM(salary) (Staff)
```

- B005: John White (30000) + Julie Lee (22000) = 52000
- B003: Ann Beech (12000) + David Ford (18000) = 30000
- B007: Mary Howe alone = 9000

Result

branchNo	SUM(salary)
B005	52000
B003	30000
B007	9000

WORKED EXAMPLE 2

Count of Staff Per Position

```
position G F COUNT(staffNo) (Staff)
```

- Managers: John White, Julie Lee — group size 2
- Assistants: Ann Beech, Mary Howe — group size 2
- Supervisor: David Ford — group size 1
- Changing the grouping attribute changes the entire shape of the result.

Result

position	COUNT(staffNo)
Manager	2
Assistant	2
Supervisor	1

Multiple Aggregates in One Grouped Query

```
branchNo G F COUNT(staffNo), AVG(salary), MAX(salary) (Staff)
```

Result

branchNo	COUNT(staffNo)	AVG(salary)	MAX(salary)
B005	2	26000	30000
B003	2	15000	18000
B007	1	9000	9000

B005's average is $(30000 + 22000)/2 = 26000$, maximum is John White's 30000. B003's average is $(12000 + 18000)/2 = 15000$. B007 has only one row, so count, average, and maximum all equal Mary Howe's single salary.

SECOND EXAMPLE

Average Rent and Count Per Property Type

PropertyForRent (propertyNo, street, city, type, rooms, rent, branchNo)

propertyNo	city	type	rent	branchNo
PA14	Islamabad	House	42000	B007
PL94	Karachi	Flat	28000	B005
PG4	Lahore	Flat	21000	B003
PG36	Lahore	Flat	26000	B003
PG21	Lahore	House	39000	B003

```
type G F COUNT(propertyNo), AVG(rent)
      (PropertyForRent)
```

Result

type	COUNT	AVG(rent)
House	2	40500
Flat	3	25000

Flat: $28000 + 21000 + 26000 = 75000 \div 3 = 25000$. House: $42000 + 39000 = 81000 \div 2 = 40500$.

Combining Grouping With Selection

“Find the total rent collected on properties managed by branch B003”

σ branchNo = 'B003'
Filter first: keep PG4, PG36, PG21



$\mathcal{F}$ SUM(rent)
21000+26000+39000 = 86000

WARNING

Filter with σ before grouping whenever the condition concerns individual tuples. branchNo $\mathcal{G}$ $\mathcal{F}$ SUM(rent)(PropertyForRent) and type $\mathcal{G}$ $\mathcal{F}$ SUM(rent)(PropertyForRent) are both valid but answer completely different questions — state the question in plain English before writing the algebra.

KEY TAKEAWAYS

Aggregation and Grouping

1

COUNT, SUM, AVG, MIN, MAX reduce many values from one column into a single summary value.

2

The aggregate operator $\mathcal{F}$ collapses an entire relation into one result tuple when no grouping attribute is given.

3

Grouping ($\mathcal{G}$) partitions a relation into buckets sharing a grouping-attribute value — by itself it computes nothing.

4

grouping-attrs $\mathcal{G}$ $\mathcal{F}$ function-list (R) applies aggregates separately per group, one result tuple per group.

5

Filter with σ before grouping; choosing the wrong grouping attribute is a silent logic error, not a syntax error.